



NITI Aayog



SECTOR AT A GLANCE 06

Hydropower in India

Energy in India

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HYDROPOWER IN INDIA

ENERGY IN INDIA

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Hydropower is one of India's biggest renewable energy sources. It involves energy generation via small hydropower plants, with a capacity of less than 25 MW, and large hydropower plants exceeding 25 MW in capacity. India has added about 6.38 GW of hydropower capacity since 2014, with large and small hydropower contributing ~11% and 1.16% to India's total energy mix in 2023.¹ Large hydropower's share in India's energy mix has dropped from about ~16% in 2014 to ~11% in 2023 due to rising environmental concerns and growth in coal-based power generation.²

India possesses a substantial hydropower potential of approximately 145 GW, with an additional 119 GW potential in pumped hydro projects and 21.113 GW in small hydro potential, collectively constituting a significant source for meeting the country's energy security requirements.³ Further, hydropower can be a critical source to provide reliable electricity in remote and hilly areas of the country, ultimately promoting long-term energy security in India.

However, challenges such as low private sector participation (less than 10% in hydropower generation compared to over 90% in renewable energy sources), the slow pace of environmental clearances, lack of reliable hydrological data, high costs of power evacuation in remote locations, and social resettlement issues in large hydropower projects have impeded the timely and efficient development of hydroelectric projects.⁴

Given these opportunities and challenges, financial support for enabling infrastructure support, promotion of hydropower R&D, and waiver of Inter-State Transmission System (ISTS) charges are key initiatives taken by Government of India. Several ongoing and recent central government initiatives, including but not limited to declaring large hydropower projects as part of non-solar RPOs, Jal-Urjamitra Skill Development Programme, focus on expanding hydropower in the country.

Efforts to upgrade and modernize infrastructure, facilitate private investment, and promote adequate environment safeguards and community engagement will be critical to continue advancing India's hydropower ambitions. This note presents key trends related to different components of hydropower in India and actions for the way forward.

Figure 1: Key hydropower statistics

162.16 TWh
Electricity supply from Hydro in 2020

234.56 TWh
Electricity supply from Hydro in 2047

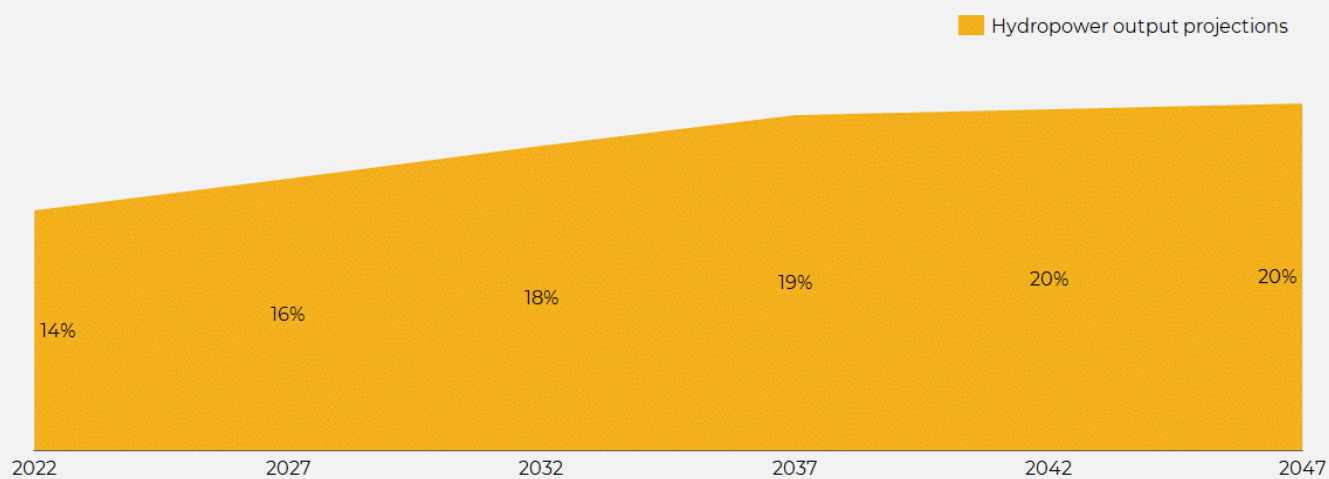
12.12%
Hydro energy contribution in India's energy mix in Dec 2023

67.77%
Unexploited large hydro potential in India in Dec 2023

Source: Central Electricity Authority, Installed Capacity Report, 2023; NITI Aayog, India Energy Security Scenario, accessed December 2023; PIB, Utilization of Hydro Power Potential - 42104.6 MW (29%) out of 145320 MW developed and 15023.5 MW (10.3%) is under construction, 2023

1 Ministry of Power, Power Sector at a Glance, 2023; International Energy Agency, India Energy Outlook, 2021; Central Electricity Authority, Installed Capacity Report, 2023
2 Ministry of Power, Growth of Electricity Sector in India From 1947-2020, 2020; Observer Research Foundation, Hydropower in India: Balancing global carbon benefits with local environmental costs, 2022; Central Electricity Authority, Installed Capacity Report, 2023; Central Electricity Authority, Installed Capacity Report, 2023
3 PIB, Utilization of Hydro Power Potential - 42104.6 MW (29%) out of 145320 MW developed and 15023.5 MW (10.3%) is under construction, 2023; PIB, Speeding up Hydro Power: Central Electricity Authority Fast-tracks Approval Mechanism for Pumped Storage Projects, 2023; MNRE, Small Hydro Energy, accessed July 2023
4 National Power Portal, All India Installed Capacity of Power Stations, 2023

Figure 2: Projected growth of Hydropower 2022-2047



Source: NITI Aayog, India Energy Security Scenario, accessed December 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by increasing the adoption of hydropower. These include providing energy supply in hilly and remote regions, ensuring energy security, and increasing energy storage capacity.



Ensuring energy security:

Hydropower plants offer the advantage of rapid and cost-effective startup and shutdown, providing crucial flexibility to network operators in addressing significant variations in electricity demand throughout seasons and different times of the day.⁵ Although hydropower plants require substantial initial investments, they have extended and productive lifespans that significantly reduce costs over time. The Bhakra Nangal plant, for instance, has meagre operating costs at just INR 0.10 per unit.⁶



Providing RE supply in remote areas:

Watermills are India's most popular form of SHP projects. Connected to small water sources such as streams or rivers, they generate electricity in off-grid areas and are often used to power small machines used in rice hulling and grain mills. SHP projects are best suited for decentralised energy generation in hilly states like Uttarakhand and Himachal Pradesh and many remote regions in Northeast India.⁷



Co-benefits of hydropower dams in flood control and climate change adaptation:

There is global enthusiasm for hydropower fuelled by its multiple advantages, including: 1) addressing the challenge of intermittent energy supply resulting from the growing use of renewable energy sources in the grid and significantly reducing global carbon emissions, 2) adapting to hydrographic flow patterns influenced by increased precipitation and potential extreme rainfall events, and 3) meeting additional water requirements for agricultural irrigation due to rising temperatures.⁸



Utilising PSPs as a flexible energy generation and storage asset:

Due to the intermittency of variable renewable energy sources, there is a growing demand for flexible energy generation and storage assets. PSPs are well-suited for this as they provide large-scale storage with a service life of over 40-50 years, surpassing that of other storage technologies, and are environmentally friendly and non-polluting. In recognition of its role, 39 Hydro PSPs of 47 GW are being pursued to be commissioned by the year 2030.⁹

⁵ World Bank, India Hydropower Development, 2012

⁶ World Bank, India Hydropower Development, 2012

⁷ Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

⁸ Observer Research Foundation, Hydropower in India: Balancing global carbon benefits with local environmental costs, 2022

⁹ PIB, Speeding up Hydro Power: Central Electricity Authority Fast-tracks Approval Mechanism for Pumped Storage Projects, 2023



Hydropower contributes to green jobs, especially in remote areas

Hydropower sector is one of the largest employers among clean energy sources employing people across manufacturing, construction and installation, and O&M services. India accounts for 18% of the global hydropower employment (direct jobs) amounting to more than 400,000 jobs.¹⁰ Particularly in the hilly and remote regions, small hydropower plants create jobs and boost local economies.¹¹ As the capacity of hydropower output is set to increase by more than 50% by 2047, there is huge potential of employment generation.¹²

Challenges

This section highlights some of the major challenges and constraints facing hydropower in India. Efforts to resolve such challenges will accelerate the installation of small and large hydropower dams, reduce cost of power evacuation in remote locations, and broadly smoothen India's clean energy transition journey.



Pace of environmental clearances:

Delays and lengthy approval processes to receive green clearances delay hydropower projects. As a 2019 report by the Standing Committee on Energy on hydropower finds, the need for environmental, forest, and wildlife clearances from three disparate wings of the central environment ministry can be streamlined, while river basin-based studies prior to project planning can support more informed decision-making as part of the clearance process.¹³



Limited feasibility assessments:

The lack of reliable hydrological data in India makes it difficult to assess the financial and technical feasibility of small hydropower (SHP) projects in different regions. Insufficient or inaccurate data can lead to flawed design parameters and potentially cause financial losses throughout the project's lifespan.¹⁴



High cost of power evacuation in remote locations:

SHPs are often located in remote areas without proper evacuation infrastructure. The cost of extending transmission lines exclusively for small hydropower projects is high due to their small scale. Moreover, relying on low-capacity rural networks for power evacuation adds financial burdens for small hydropower developers.¹⁵



Infrastructure-related risks:

SHP projects in remote hilly regions face additional risks, such as natural disasters like landslides and limited road connectivity.¹⁶ These challenges also increase the likelihood of surpassing the allocated budget, potentially resulting in cost overruns. India has many ageing dams that are structurally vulnerable and require renovation and modernisation (R&M). Around 619 large dams had already surpassed the 50-year mark by 2015, and about 64 dams in India are over 115 years old.¹⁷

¹⁰ IRENA, Renewable energy and jobs, 2022

¹¹ US Department of Energy, Benefits of Hydropower, accessed on December 2023

¹² NITI Aayog, India Energy Security Scenario, accessed on December 2023

¹³ Standing Committee on Energy, Forty-Third Report on Hydro-power relating to the Ministry of Power, 2019

¹⁴ Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

¹⁵ Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

¹⁶ Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

¹⁷ India Water Portal, Dams, in distress?, 2019



Environmental challenges and social resettlement issues in large hydropower projects:

Disrupting the natural water flow causes significant harm to downstream agricultural activities and affects the delicate balance of wildlife and habitat ecosystems due to the loss of hydrological connectivity.¹⁸ Twelve projects with a combined capacity exceeding 3,500 MW have been suspended or abandoned due to environmental concerns as of 2021. Moreover, opposition from local communities, driven by worries about the ecological consequences, has caused delays or complete abandonment of forty projects with a capacity of 13,633 MW.¹⁹ As the 2019 report by the Standing Committee on Energy on hydropower observes, land acquisition for hydropower projects is growing more uncertain and time-consuming, and recommends that district administrations (supported by the central and state governments and project developers) ensure transparent and fair rehabilitation, resettlement, and compensation for affected communities.²⁰



High Taxation on PSP components:

The concessions that are available to other renewable sources are yet to be extended to pumped storage projects. Currently, PSP components are subject to a GST rate of 18% to 28%, whereas solar and wind energy components are taxed at a lower rate of 12%, which affects the viability of PSPs.²¹



Risk in the reservoir management due to extreme climate change events:

Climate change impacts on hydropower potential vary by region, with anticipated increases in North India due to elevated glacial melting and in Central India owing to higher rainfall. Conversely, some hydropower dams in South India may experience a decline in potential due to increased evapotranspiration. The projected rise in extreme inflow events poses challenges in reservoir management for flood control, hydropower production, and water supply. Additionally, oversupply to reservoirs heightens the risk of dam breaks.²²

¹⁸ Energy Alternatives India, India Hydro Energy, Accessed June 2023; TERI, Green Growth and Hydro Power in India, 2015

¹⁹ Observer Research Foundation, Hydropower in India: Balancing global carbon benefits with local environmental costs, 2022

²⁰ Standing Committee on Energy, Forty-Third Report on Hydro-power relating to the Ministry of Power, 2019

²¹ Ministry of Power, Draft guidelines to promote development of Pump Storage Projects (PSP) in the country – seeking comments - regarding, 2023

²² Dipesh Singh Chuphal, Vimal Mishra, Increased hydropower but with an elevated risk of reservoir operations in India under the warming climate, iScience, Volume 26, Issue 2, 2023

Institutional framework and policy landscape —

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee hydropower energy.



NITI Aayog:

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. For hydropower, it conducts feasibility assessments of projects, infrastructure evaluation, policy analysis, and more to promote efficient hydro power utilization.



Ministry of Power:

The ministry is responsible for all matters related to large hydropower projects. It is actively involved in research, development and providing technical assistance relating to hydro-electric power.



Ministry of New and Renewable Energy (MNRE):

The ministry is vested with the responsibility of promoting and developing small hydropower plants. It supports the sector through resource assessment, detailed survey and investigation, capital subsidy for state sector projects, research and development, and human resource development, including the strengthening of technical institutions and capacity building.

Major policies and programs

This section highlights existing policies related to integrated energy assessments and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



2019 hydropower measures:

In 2019, the Union Cabinet approved several measures to promote India's hydropower sector. These measures included declaring large hydropower projects (those greater than 25 MW) as a renewable energy source (previously limited to projects less than 25 MW), introducing tariff rationalization measures, providing budgetary support for enabling infrastructure (e.g., roads and bridges) and flood moderation components. Additionally, large hydropower projects (LHPs) were declared a part of non-solar renewable purchase obligations. The Hydropower Purchase Obligation (HPO) was created as a separate entity within the non-solar renewable purchase obligation. This was followed by the introduction of necessary amendments in Tariff Policy and Tariff Regulations to operationalize HPO by all states.²³

²³ Cabinet, Cabinet approves Measures to promote Hydro Power Sector, 2019

Electricity Act, 2003:

The Act was critical for the hydropower sector, as it eliminated the requirement for mandatory techno-economic clearance from the Central Electricity Authority (CEA), except for hydroelectric schemes with exceptionally high capital costs. The Act also facilitated the implementation of Open Access, established provisions for trading, and promoted the development of hydropower markets.²⁴

Resource assessment programs:

The Ministry of New and Renewable Energy financially supports resource assessment activities such as site identification, preparation of DPR, site surveys, investigations etc. It provides INR 6 lakh for resource assessments up to 1 MW station capacity and INR 10 lakh for more than 1 MW station capacity to expand SHP in India.²⁵

Renovation programs:

The MNRE has sought to renovate and modernise many of India's old SHP projects by offering financial incentives to the tune of INR 10,000/kW for projects up to 1000 KW and INR 1 crore/MW for projects up to 25 MW to all states and Union Territories.²⁶

R&D and Jal-Urjamitra Skill Development Programme:

The MNRE is providing financial support to states conducting hydropower R&D and covering up to 100% of the capacity-building costs required for improving the technical capabilities of the hydropower workforce.²⁷ Through the Jal-Urjamitra Skill Development Programme, it is training technicians to install, operate, repair and maintain small hydro power plant projects.²⁸

Waiver of Inter-State Transmission System (ISTS) charges for new hydro projects:

In line with the National Tariff Policy 2016 and to facilitate renewable energy grid integration, the Ministry of Power waived Inter-State Transmission System and other transmission charges to Pumped Storage Projects. This move is anticipated to boost the hydro sector, enhance India's water security, and promote development in hilly states, where a significant portion of hydro potential is concentrated.²⁹

24 Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

25 Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

26 Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

27 Ministry of New & Renewable Energy (MNRE), Evaluation of Small Hydro Power (SHP) Programme of MNRE, 2017

28 PIB, Steps by Government of India to boost employment opportunities in the green energy sector , 2022

29 PIB, Ministry of Power issues order for waiver of ISTS charges on transmission of electricity generated from new hydro-power projects as a further step to realise the Government's commitment to achieve its power requirement from renewable energy sources, 2022

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the ASSOCHAM and the Observer Research Foundation.

 Strengthen infrastructure resiliency and efficiency	Upgrade and modernize infrastructure and generation projects - Develop the enabling infrastructure for investment in mega hydropower projects in line with financial, environmental, and socio-economic viability - Establish dedicated hydropower evacuation infrastructure (e.g. through the Green Energy corridor) for projects in remote sites	Invest in critical R&D - Invest in R&D into advanced hydropower technologies, such as pumped storage, small, and mini hydropower projects (e.g., state-level Alternate Hydro Energy Centres such as that established at the Indian Institute of Technology, Roorkee)
 Increase private sector engagement	Create an enabling regulatory environment - Streamline project clearance, approval, allocation, and land acquisition processes at the centre and state government level, with dedicated agencies to address delays and coordination issues (including inter-state dispute resolution)	Reintroduce and expand scope of fiscal incentives - Increase developer access to long tenure debt and other forms of concessional finance, in addition to customs exemptions and export benefits - Encourage innovative financial instruments and sources, such as international climate finance
 Manage environmental footprint and promote community engagement	Enhance project planning and safeguards for affected stakeholders by through better environmental footprint management - Support states with integrated river basin studies and planning to ensure optimum project development	Provide community incentives and support transparent and participatory engagement - Enable and promote transparent and participatory social impact assessments, with sufficient public involvement in consultative project planning and decision-making processes - Ensure developers provide sufficient skill development and employment opportunities (in addition to the development of public infrastructure) to enhance social acceptance for projects

Recommended readings

ORF. (2023). *Hydropower in India: An update*.

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